Department of Computer Science and Engineering
Academic Year: 2025-2026 (Even Semester)
INNOVATIVE TEACHING PRACTICES

Degree, Semester & Branch: IV Semester B.E. CSE
Course Code & Title: CS3452 & Theory of Computation
Name of the Faculty member: Dr. S. Poornam
Date & Time: 21.01.2026 & 11:50 A.M
Topic: Minimization of DFA
Course Outcome: CO1
Type of Activity: Peer-Led Learning Approach

Justification

This peer-led learning activity helps students understand the concept of DFA minimization through collaborative learning, group discussion, and peer explanation. DFA minimization is often difficult for students due to the multiple steps involved in identifying equivalent states and constructing the minimized DFA. By allowing students to work in teams and explain concepts to one another, the activity promotes active learning, improves analytical thinking, and enhances conceptual clarity. The approach encourages students to learn from peers, discuss different problem-solving strategies, and build confidence in applying minimization techniques.

Activity Description

Peer-Led Learning Activity: Minimization of DFA

Introduction Phase

- The instructor briefly introduced the importance of DFA minimization in reducing the number of states and improving automata efficiency.
- Students were divided into small peer groups.
- Each group was assigned a DFA minimization problem.

Peer Learning and Discussion Phase

- Students identified reachable and unreachable states in the given DFA.
- Group members collaboratively prepared state equivalence tables.
- Peer leaders in each group guided the discussion on:
    - Distinguishing states,
    - Equivalent states,
    - Partitioning method,
    - Construction of minimized DFA.
- Students explained each minimization step to fellow group members.
- The instructor monitored the discussions and clarified doubts when required.

Presentation Phase

- Each group presented:
    - The original DFA,
    - Steps followed in minimization,
    - Final minimized DFA.
- Peer groups compared different approaches and discussed common mistakes in minimization.

Assessment and Reflection Phase

- Students solved an additional DFA minimization problem individually.
- A short reflective discussion was conducted on:
    - Challenges faced during minimization,
    - Benefits of peer learning,
    - Applications of DFA minimization in compiler design and pattern matching.

CO – PO / PSO Mapping

CO	PO1	PO2	PO3	PO4	PSO1	PSO2	PSO3
CO1	2	2	2	1	2	1	2

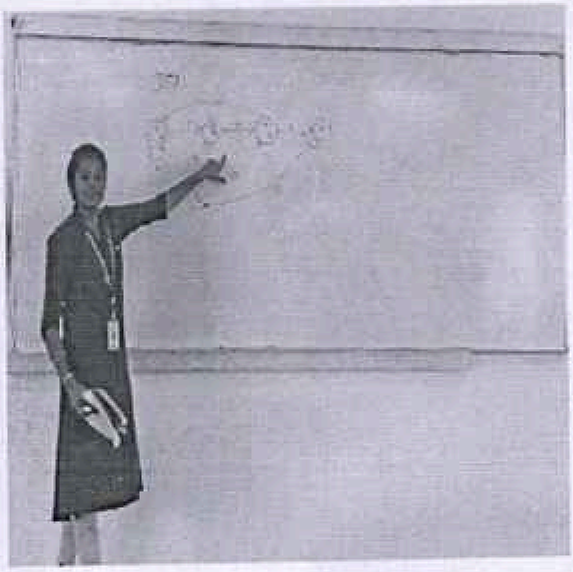
(1 – Low, 2 – Moderate, 3 – High)

PO/PSO mapped: PO1, PO2, PO3, PO4, PSO1, PSO2, PSO3

Justification for correlation

- Students applied theoretical concepts of automata to solve DFA minimization problems.
- Peer discussion enhanced analytical and problem-solving abilities.
- Collaborative learning improved communication and teamwork skills.
- The activity strengthened understanding of equivalence partitioning and state reduction techniques.
- Students developed confidence in explaining formal language concepts to peers.

Images / Screenshot of the practice:



Outcome

- Students successfully identified equivalent and distinguishable states in DFA.
- They learned to construct minimized DFA systematically.
- Peer-led discussions improved participation and conceptual understanding.
- Students gained confidence in solving automata-related problems independently.
- The activity promoted collaborative and interactive learning.

Reflective Report

Pre-Implementation

Identified Challenges:

- Students found DFA minimization procedures lengthy and confusing.
- Difficulty in understanding equivalence partitioning techniques.
- Limited confidence in solving minimization problems individually

Addressing Challenges:

- Introduced peer-led collaborative learning strategies.
- Encouraged group discussions and peer explanations.
- Provided step-by-step DFA examples for guided practice.

Post-Implementation

- Students actively participated in group problem-solving activities.
- Peer interaction helped clarify misconceptions quickly.
- Most students were able to minimize DFA independently after the session.
- Group presentations improved communication and technical explanation skills.
- Feedback indicated that peer learning made Theory of Computation concepts easier and more engaging.

References

- https://www.geeksforgeeks.org/minimization-of-dfa/
- https://www.tutorialspoint.com/automata_theory/dfa_minimization.htm
- AdvancED: The Institute for the Advancement of Higher Education | Vanderbilt University
- https://www.edutopia.org/article/peer-learning-strategies-classroom

7/5/26

Signature of Faculty Member

7/5/2026

HOD